

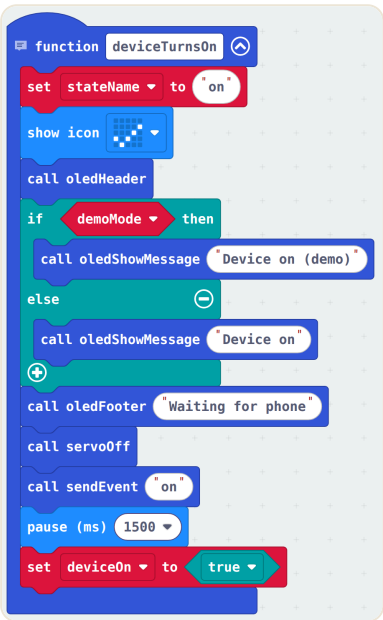
Sunburn device: the

The Sunburn Flowchart, built from the real blocks of the program

The Sunburn device is a wearable built on a BBC micro:bit, a small computer with 25 LEDs where you are through a phone, and then flashes, beeps or taps your arm until you put built from 78 coloured blocks. Every one of them is on these sheets, with a note saying

Follow the arrows. Each red label is a box of the flowchart; the coloured picture under it is the actual block forever block runs this plan round and round, all day.

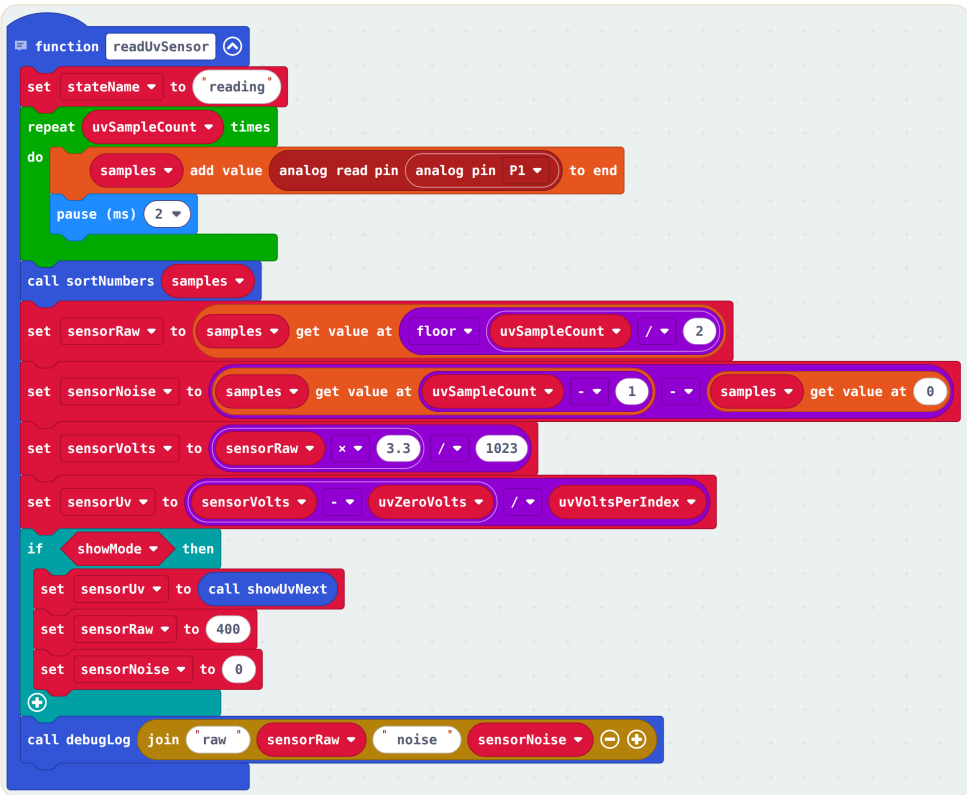
Device turns on



deviceTurnsOn

Switching on: shows a tick, says Device on on the screen, tells the app, and marks the device as on.

Read the UV sensor



readUvSensor

Reads the UV sensor plugged into pin P1. It takes 15 quick readings and uses the middle one, so a single odd reading cannot set off a false alarm. Then it turns the sensor's voltage into a UV index number. In show mode the reading is replaced by a pretend value that climbs through every band.

Is the phone sending the online UV index?

Pick the UV value: the higher of sensor and online

How str

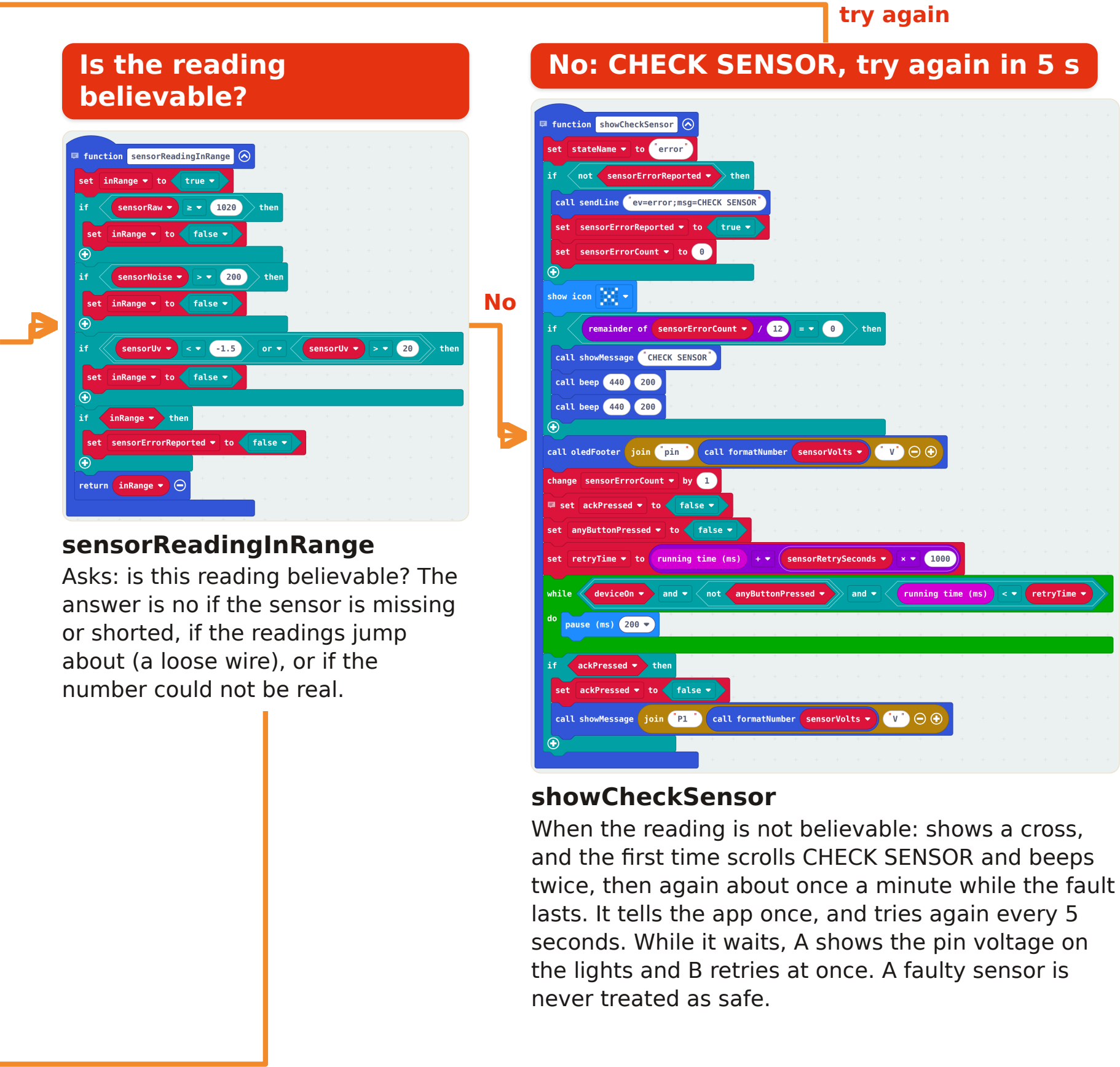
Yes or No

program on the wrist

am, with what each one does

hts and two buttons. It measures the sun's UV with a sensor, checks the online UV index for sunscreen on. Two hours later it reminds you to reapply. The program that does all this is what it does.

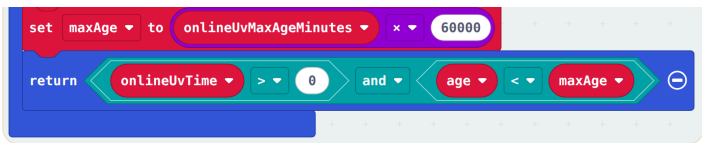
k of code that does that job, exactly as it looks in the MakeCode editor, and the note says what it does. The



ong is the sun?



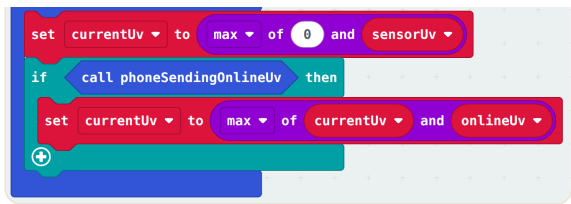
The Sunburn Flowchart



phoneSendingOnlineUv

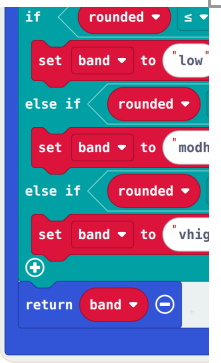
Asks: has the phone app sent an online UV index in the last 30 minutes?

Yes or No



chooseUvValue

Picks the UV number the device will act on: the higher of the sensor's reading and the phone's online value, or just the sensor's reading when the phone is not sending.



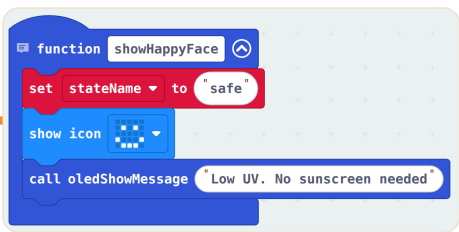
uvBand

Turns a UV index into four levels (low, moderate, high, and very high (8 and above)).

0 to 2

3 to 7

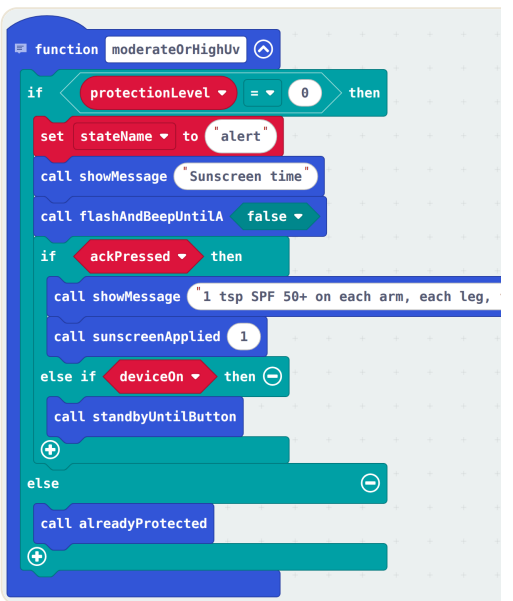
0 to 2, low: happy face



showHappyFace

UV 0 to 2: shows a happy face. No sunscreen needed right now.

3 to 7, moderate or high: flash and slow beep until A

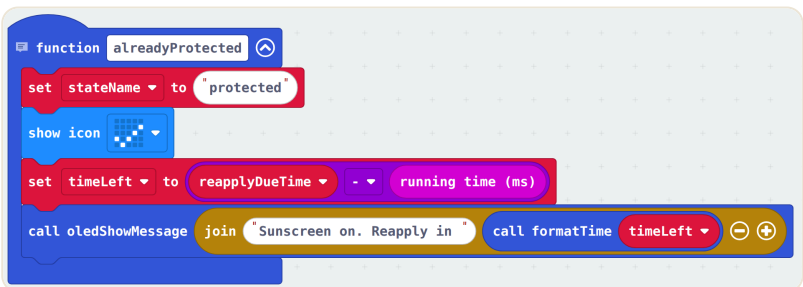


moderateOrHighUv

UV 3 to 7: flashes and beeps until A is pressed, then shows how much sunscreen to put on and starts the 2 hour timer. If sunscreen is already on, it just shows a message and the time left. If nobody presses A within 10 minutes, the device assumes it is not being worn and goes to sleep.

sunscreen already on A pressed

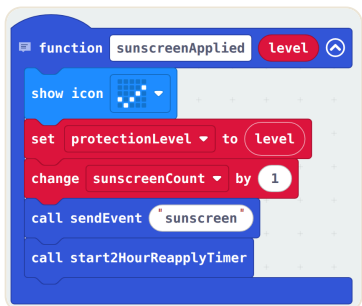
Sunscreen already on: just a tick



alreadyProtected

Sunscreen is already on, so there is no alert: just a tick and the time until it needs reapplying.

A pressed: sunscreen on



sunscreenApplied

A was pressed after a sunscreen alert: remembers which advice given, counts one more application, tells the app, and starts the timer.

then back to the top

then back to the top



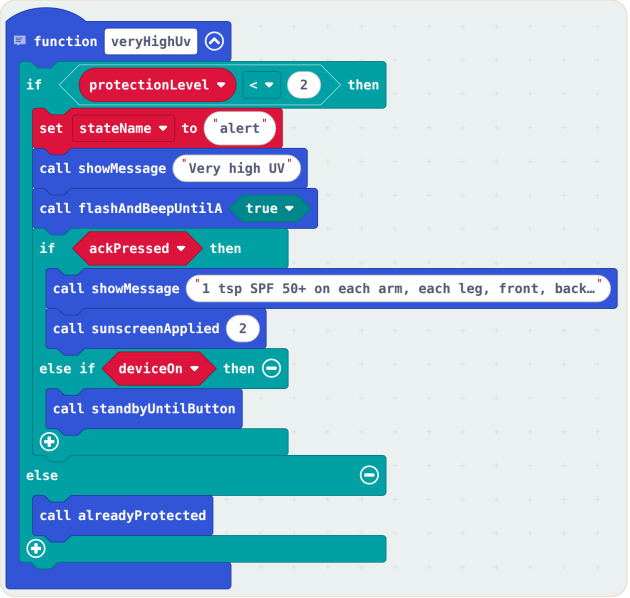
index into one of the
on the flowchart: low (0
rate or high (3 to 7),
to 10) or extreme (11

0 7 8 to 10 11 and above



s slowly
/s how
nd
nscreen
tick
presses
/ice
n and

8 to 10, very high: flash
and fast beep until A



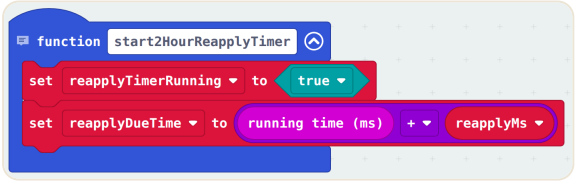
veryHighUv
UV 8 to 10: flashes and beeps
quickly until A is pressed, then
shows the sunscreen advice plus
hat and shade, and starts the 2
hour timer. It alerts again if the
wearer so far only had the
moderate advice.

A pressed

n is

Start the 2 hour reapply
timer

Sleep until any button is
pressed



start2HourReapplyTimer
Starts the 2 hour countdown to the
next sunscreen.

een
ce was
cation,
imer.

11
ta



extr
UV 1
is pre
inste
sleep

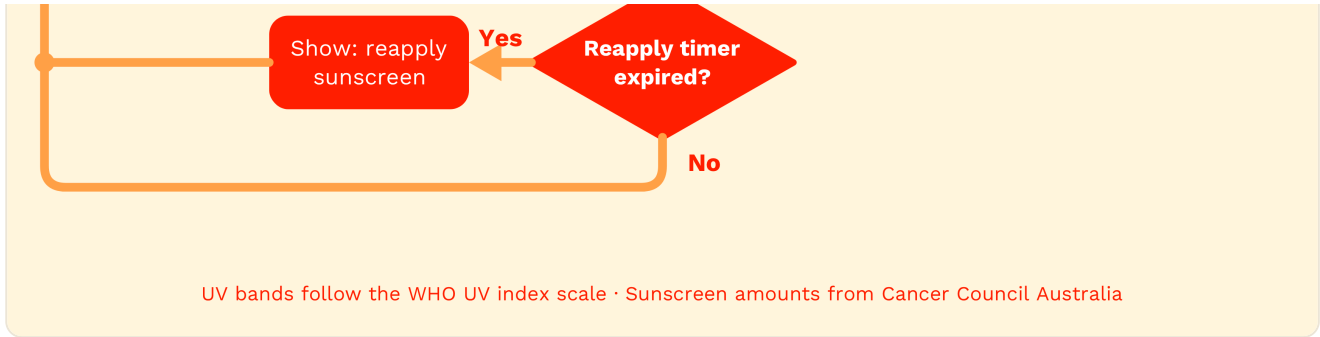
a button was pressed

and above, extreme:
to the arm until A



on extremeUv

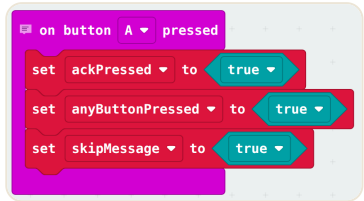
1 or more: taps the arm until A pressed (after 2 minutes it beeps and), shows Go inside now, then goes until a button is pressed.



The plan the device follows. Every box on it is a block on the left, with the same name.

The two buttons

A: done

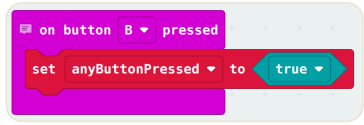


on button A pressed

Pressing A means done: sunscreen is on, or I am going inside. It also wakes the device, skips a long message, and during a wait shows the UV number and the countdown.

Pressing A and B together turns the device off, or on again (see the flowchart).

B: wake up



on button B pressed

Pressing B wakes the device from sleep. Holding B while switching on gives the fast demo timings.

The whole program



All 78 blocks in the editor, in nine sections. The flowchart on this sheet is sections 1 to 4; sections 5 to 9 are the helpers: screen, sound, Bluetooth, maths and the optional screen.

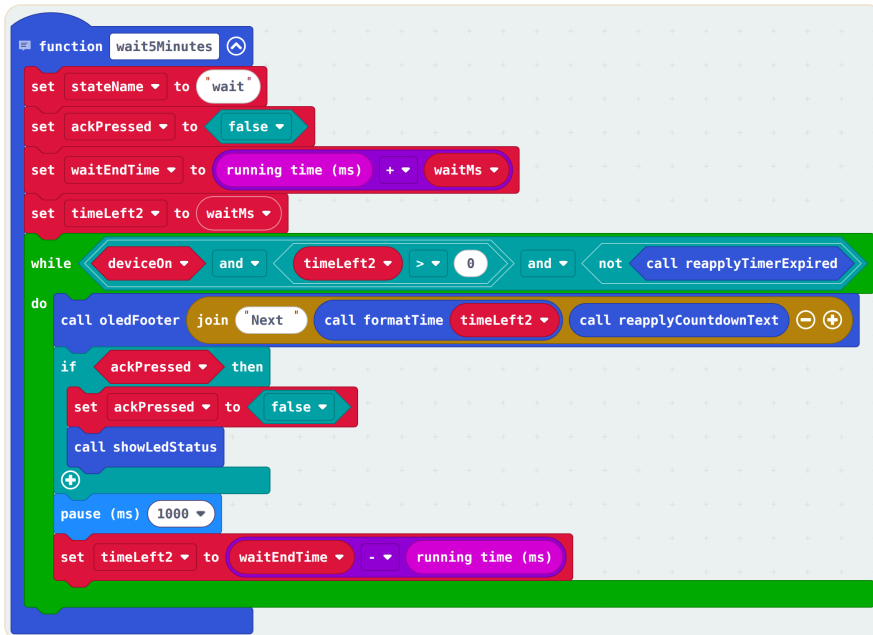
What the block colours mean

- | | |
|-------------------------------|-----------------------------|
| Variables: remember a value | Text |
| Functions: a step with a name | Arrays: lists |
| Basic: the lights, pauses | Music: beeps |
| Input: buttons and time | Pins: sensor, motor, screen |
| Loops: repeat | Bluetooth: the phone |
| Logic: if, compare | Serial: the USB cable |
| Maths | Control: timing |

Words used here

UV index How strong the sun's burning rays are, from 0 (none) to 11 and above (extreme). Protection is needed from 3 upwards, even on cool or cloudy days.

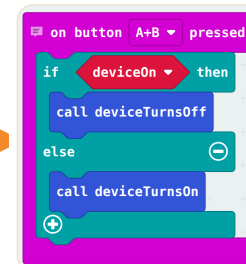
Wait 5 minutes



wait5Minutes

Waits 5 minutes before the next check. The wait ends early if the device is turned off or the sunscreen timer runs out. Pressing A during the wait shows the UV number and the countdown.

A and B held?



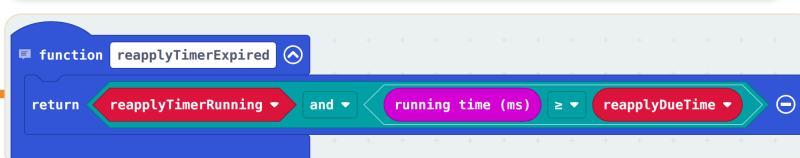
on button A+B pressed

Pressing A and B together turns the device off, or on again.

Yes

No

Is it time for more sunscreen?



reapplyTimerExpired

Asks: have the 2 hours since the last sunscreen passed?

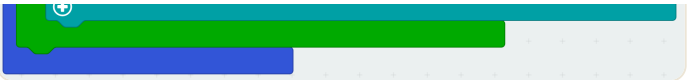
Yes

Yes: show Reapply sunscreen



showReapplySunscreen

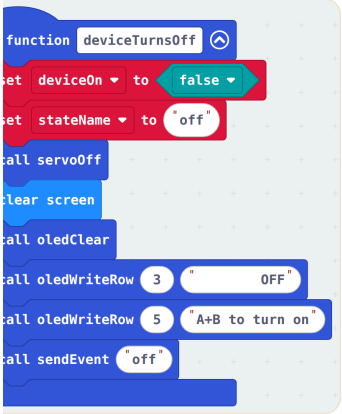
Shows Reapply sunscreen and clears the timer, so the next trip round the flowchart alerts again if the sun is still strong.



standbyUntilButton

Sleep: waits for any button, or done from the app. Pressing A and B together still turns the device off. Afterwards it goes straight back to reading the sensor. In show mode it wakes by itself after 5 seconds.

Yes: device turns off



eviceTurnsOff

Switching off: everything stops and the screen shows OFF. Bluetooth says on so the app can switch it back on.

micro:bit The small computer the program runs on. It has a 5 by 5 grid of lights, buttons A and B, and sockets called pins (P0, P1, P2) that other parts plug into.

Block One instruction of the program. Blocks snap together and run in order from top to bottom. A block with a name at the top (function) is a step that other blocks can call on.

Sensor The part that measures UV. It gives a voltage, which the program turns into a UV index number.

Servo A small motor with an arm. Here it taps the wearer's wrist when the sun is extreme.

Bluetooth and USB The two ways the wearable talks to a phone or a laptop. The phone app sends the online UV index; the wearable sends back what it is doing.

OLED An optional small screen. Without one, the grid of lights does the job.

ms and Hz ms are milliseconds, thousandths of a second. Hz sets how high a beep sounds.

Demo timings Much shorter waits (seconds instead of minutes and hours) so the whole day can be shown in a few minutes at a demo.

Names in the blocks Words like uvSampleCount are the program's own names for its settings and memory. Read them as English: uv sample count is how many readings to take.

Every one of the 78 blocks, with its note, is in the booklet on the table.